ChatGPT

Designing and Prototyping a Solid-State AI Pendant with Ternary Computing

1. Concept Overview

Massimo's goal is to create a pendant-sized AI assistant similar to the **Plaud NotePin**, but with an integrated camera. The device will take advantage of emerging **solid-state battery** technology for safer, higher-density power and will experiment with **ternary computing** to showcase a non-binary architecture. Below we outline the design choices, components and steps needed to build a working prototype.

Inspiration – Plaud NotePin

The Plaud NotePin is a compact AI voice-recording wearable. It has a capsule-like design (51 × 21 × 11 mm) weighing only 16.6 g and houses dual MEMS microphones, a 270 mAh battery and 64 GB of storage [1]. It can be worn as a necklace, bracelet, clip or pin [1]. Massimo's pendant will adopt a similar form factor but will add a front-facing camera and use ternary logic internally.

2. Key Components

Subsystem	Choice/Description	Reasons
Ternary CPU	**5500FP Balanced Ternary RISC Processor**: 24-trit word size, 120-instruction ISA, runs at ~20 MHz on an FPGA board measuring ~7.5 × 4.3 cm [2] [3].	Balanced ternary (−1, 0, +1) enables higher information density and lower switching activity; a 41-trit system is roughly equivalent to 64 bits [4]. The 5500FP board is available for research and demonstrates the feasibility of ternary computing [2].
Bridge MCU	**Low-power microcontroller (e.g., ESP32-S3)** for interfacing sensors, power management and wireless connectivity (Wi-Fi/Bluetooth).	The ternary CPU currently requires an FPGA board; therefore a binary MCU is included to handle real-time tasks, wireless protocols and to translate between ternary and binary domains.
Camera	**OV6948 micro image sensor**: world's smallest commercial sensor at 0.575 × 0.575 mm [5]; 200 × 200 resolution at 30 fps [6]; low power (~25 mW) and 4-pin interface [7].	Tiny size fits easily within the pendant; adequate for simple photos or video.

Subsystem	Choice/Description	Reasons
Microphones	**Dual MEMS microphones** (similar to Plaud NotePin).	Provide stereo audio capture for AI transcription and ambient understanding.
Memory & Storage	64 GB eMMC or microSD, plus 8–16 MB PSRAM for buffering.	Matches Plaud NotePin's capacity [1] and allows local caching of audio/video data and AI embeddings.
Solid-State Battery	**Flexible solid-state battery (~300 mAh)**. Solid-state cells replace liquid electrolytes with ceramic/polymer materials [8]. They offer 2–3× higher energy density, charge in 3–12 minutes and have enhanced safety (reduced fire risk) [9]. Lifespan can exceed 2 000–10 000 cycles [10].	Allows a safer, high-capacity power source in a thin form factor. The battery's stability reduces leakage and thermal runaway [9] —crucial for a wearable.
User Interface	**Tactile button** on the right side; RGB status LED on front/top.	Provides simple user interactions: one-press to record, double-press to capture a photo; the LED indicates power, recording status or alerts.
Enclosure	3D-printed bioplastic or aluminum case; IP54 splash-resistant.	Protects electronics and contributes to the premium feel.

3. System Architecture

3.1 Hardware Block Diagram

- **Ternary Processor (5500FP)**: handles high-level AI inference tasks that benefit from ternary logic's efficiency. It communicates with the MCU over a UART/SPI bridge. Because the 5500FP uses balanced ternary, each trit can represent −1, 0 or +1, giving higher information density and lower switching activity compared to binary [11].
- **Microcontroller (MCU)**: orchestrates peripherals (camera, microphones, storage, battery management) and manages wireless communication. It also acts as a fallback compute core when ternary processing is unavailable.
- **Camera & Audio**: the OV6948 sensor feeds analog video to an on-board ADC/ISP module; audio from MEMS mics enters via I^2S. Data is packetized and streamed to the MCU.
- **Power Management**: the solid-state battery connects to a power-management IC providing 3.3 V and 1.2 V rails. Fast-charge circuitry leverages the battery's ability to recharge in minutes [9].

3.2 Software Architecture

- **Firmware (MCU)**: manages boot-strapping, sensor drivers, file system, BLE/Wi-Fi stack and secure communication with a smartphone app or cloud service.
- **Ternary OS Layer**: a minimal operating layer on the 5500FP executing AI inference tasks. The 120-instruction balanced ternary ISA [2] can be targeted via a cross-compiler (C or a custom **Treal** dialect). Weights for models (e.g., small transformer for speech summarization) are quantized to

ternary values (−1, 0, +1), reducing memory footprint and leveraging the ternary processor's efficiency [12].

- **AI Capabilities**:
- **Speech capture & summarization**: audio is transcribed to text on-device or via an edge server. Summaries are generated using quantized models to fit the ternary hardware.
- **Contextual image capture**: the camera captures 200 × 200 frames; the ternary processor can run lightweight vision algorithms (e.g., object detection or scene classification) for context-aware annotations.
- **Assistant logic**: voice commands control recording, summarization and retrieval. Haptic alerts (vibration) notify the user of important events.
- **Companion App**: a smartphone app syncs data, displays captured notes/images and acts as a UI for settings. It also updates firmware over the air.

4. Mechanical and Industrial Design

1. **Form Factor**: follow Plaud NotePin's dimensions – roughly a 5.1 cm tall, 2.1 cm wide and 1.1 cm thick capsule [1]. The camera lens is centred on the front with a small LED above it. The right side houses the tactile button; the left side remains smooth. The backside includes magnetic contacts for charging and data transfer.
2. **Material**: use lightweight aluminum or durable bioplastic. Internal walls should isolate the camera and microphones from the battery to minimize interference.
3. **Assembly**: the ternary CPU board (7.5 × 4.3 cm [3]) won't fit inside the pendant; mount it externally during early prototyping. For a compact version, design a custom smaller ternary ASIC or embed the CPU on a flex PCB. In the meantime, use a ribbon cable to connect the pendant to the 5500FP board housed in a pocket module.
4. **Solid-State Battery Placement**: the battery can be a flat flexible pouch adhered to the back panel. Provide thermal pads to dissipate heat even though solid-state cells generate less heat [9].

5. Prototype Assembly Steps

1. **Acquire Components**: purchase the 5500FP development board [3], an ESP32-S3 module, an OV6948 camera module, MEMS microphones, solid-state battery (e.g., 300 mAh from ITEN or ProLogium), and necessary ICs (power management, ADC, storage).
2. **Design the PCB**: use a PCB design tool (KiCad/Altium) to create a flex-PCB the size of the pendant that routes camera signals, microphone traces, battery leads, button and LED. Include connectors to the MCU and to a flat cable that plugs into the external 5500FP board.
3. **3D-Print the Case**: design a shell in CAD with cavities for the camera lens, button, LED and connectors. Print prototypes in resin or nylon; iterate for fit.

4. **Assemble Hardware**:

5. Mount the camera behind the front aperture with lens flush to the exterior.

6. Solder microphones, LED and button to the flex-PCB. Attach the solid-state battery using adhesive or a bracket.
7. Connect the flex-PCB to the MCU module and attach the external ribbon to the 5500FP board. Test basic power-on and verify voltage rails.

8. **Flash Firmware**: program the MCU with firmware to handle button presses, sensor sampling and wireless communication. Load a minimal ternary OS and AI model onto the 5500FP. Use cross-compilers or an emulator to generate ternary machine code.

9. **Test & Iterate**:

10. Verify that pressing the button starts/stops audio/video capture. Check LED indications.

11. Measure battery life. Solid-state cells provide higher energy density and fast charging [9]; ensure the charging circuit limits current to specifications.
12. Evaluate AI performance. Balanced ternary processing should reduce switching activity and improve energy efficiency [11]; compare with binary MCU results.
13. **Miniaturization**: once the concept works, explore custom ASIC design or a smaller FPGA to integrate the ternary core into the pendant. Over time, the external board can be removed.

6. Challenges and Considerations

- **Availability of Ternary Hardware**: the 5500FP is experimental. Balanced ternary CPUs are still rare [2]. A working prototype may initially use the ternary board externally.
- **Software Tooling**: compilers, debuggers and operating systems for ternary processors are nascent. Expect to spend time porting models and writing low-level drivers.
- **Power & Heat**: while solid-state batteries reduce fire risk and handle more charge cycles [9] [10], ensure thermal management. Balanced ternary logic has lower switching activity [11], helping reduce heat.
- **Camera Resolution**: the OV6948's resolution (200 × 200) is adequate for context detection but not detailed photos [13]. For higher-resolution images, consider a larger sensor (e.g., 2 MP Arducam) and adjust the lens opening accordingly.

7. Future Enhancements

- **Custom Ternary SoC**: design a purpose-built ternary processor in a smaller package, enabling full integration inside the pendant.
- **Advanced AI Models**: explore larger ternary-optimized models (e.g., quantized vision transformers or whisper-like speech models) once tooling matures.
- **Modular Accessories**: create clip-on modules (heart-rate sensor, environmental monitors) that communicate via the MCU.
- **Expanded Display**: integrate a tiny OLED or LED matrix on the front for notifications.

Illustration: The following conceptual image shows a sleek AI pendant with a front camera, side button and subtle LED, hinting at ternary computing through triangular motifs.

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1 PLAUD NotePin
https://askjan.org/products/PLAUD-NotePin.cfm

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